Day 3 — Structured Reasoning in LLMs: KG-Infused Fine-Tuning & a Hallucination Proposal

LLM Research Log | Souvik Pal

Mentor: Dinesh Sir • Research Area: Large Language Models (LLMs)

Structured Reasoning in LLMs: Knowledge Graph-Infused Fine-Tuning, and a Proposal to Reduce Hallucinations

Based on: "Knowledge Graph-Infused Fine-Tuning for Structured Reasoning in Large Language Models" (Zhang et al.)

Objective

To study how fine-tuning an LLM with injected knowledge graph information improves structured reasoning, and to identify where this approach could be extended specifically toward reducing hallucination — which the original paper does not directly measure.

Problem Addressed by the Paper

The paper identifies two core weaknesses in LLMs: missing reasoning chains (the model can't logically connect separate facts) and weak entity-level understanding (it doesn't deeply grasp what an entity is or how it relates to others). Its claim is that injecting structured knowledge graphs during fine-tuning addresses both.

Methodology (Original Paper)

- A **Graph Neural Network (GCN/R-GCN)** encodes entities and relations from a KG into vector representations
- A **gating mechanism** dynamically balances reliance on plain-text understanding versus structured graph knowledge
- A **joint loss function** trains on the main task (e.g. QA) alongside an alignment objective keeping text and graph representations consistent
- A **knowledge-aware attention mechanism** lets each word in context attend to related entities in the graph

Dataset used: T-REx — a large-scale Wikipedia-derived KG dataset with millions of entity-relation-entity triples aligned to natural language sentences, filtered into smaller, semantically dense subgraphs for training.

Results (Original Paper)

The proposed method outperformed three baselines (KGLM, DRAGON, KG-SFT) — **86.4% QA accuracy**, **82.1% F1-score**, **29.7 BLEU**. A moderate learning rate (1e-4) with high subgraph coverage (90%+) gave the most stable results.

What I Learned

- A dataset for structured reasoning must contain organized facts, not just raw text
- Knowledge graphs let an LLM "check" its reasoning against verified facts, instead of relying purely on learned language patterns
- Fusing text and graph structure isn't trivial — it needs dedicated mechanisms (gating, joint loss); naively combining them doesn't work well
- Learning rate and KG coverage both meaningfully affect stability and accuracy
- **The key gap:** the paper only used one dataset (T-REx) and never tested hallucination directly — it measured task accuracy (QA-Acc, F1, BLEU), not factual reliability. This is the gap I'm proposing to address.

My Research Focus: Structured Reasoning to Reduce Hallucination

Building on this paper, my own research angle redirects the framework toward a different goal: not just "does the model perform the task well" but "does the model tell the truth, and can we measure that directly."

Datasets I'm Proposing to Use

Dataset	Role	Purpose
Wikidata5M	Knowledge source	Larger, cleaner replacement for T-REx
FB15k-237	Cross-check benchmark	Standard KG embedding benchmark for generalization
HotpotQA	Multi-hop reasoning	Tests whether the model can chain multiple facts
ComplexWebQuestions	Multi-hop reasoning	Multi-hop QA built specifically over KGs
TruthfulQA	Hallucination test	Measures false-but-plausible answers
HaluEval	Hallucination test	Benchmarks hallucination across QA, dialogue, summarization
FEVER	Fact verification	Checks whether generated claims are evidence-supported

Why more than one dataset is needed: T-REx/Wikidata5M supply the structured facts to reason over; HotpotQA/ComplexWebQuestions test multi-hop chaining; TruthfulQA/HaluEval directly test truthfulness (missing from the original paper, and the core of my motivation); FEVER adds an evidence verification layer.

Modifications I'm Proposing to the Original Method

- **Upgrade the knowledge source** — replace T-REx with Wikidata5M for larger, cleaner, more current coverage, keeping T-REx as a comparison baseline
- **Add a hallucination-specific evaluation stage** — introduce TruthfulQA and HaluEval, since the original paper never measures hallucination directly
- **Add a fact-verification loss term** — extend the joint loss ($L\_total = L\_task + \lambda L\_align$) with a FEVER-style verification term that penalizes unsupported claims
- **Test multi-hop reasoning explicitly** — use HotpotQA/ComplexWebQuestions to check whether the knowledge-aware attention mechanism can chain multiple facts, not just retrieve single ones
- **Report a hallucination rate metric** — alongside QA-Acc, F1, and BLEU, report the percentage of factually unsupported statements as a primary metric, since this most directly reflects my research motivation

Expected Outcome

By combining a stronger knowledge graph (Wikidata5M), multi-hop reasoning benchmarks (HotpotQA), and hallucination-specific evaluation (TruthfulQA, HaluEval, FEVER), I expect to show that structured reasoning not only improves task accuracy — as the original paper demonstrated — but also measurably reduces hallucination rate, the gap the original work left unaddressed.

Conclusion

The original paper gave me a strong technical foundation for how knowledge graphs are encoded, fused with language models, and jointly trained. My contribution is to redirect this framework toward a specific, measurable goal — reducing hallucination — by introducing hallucination-focused datasets and an additional verification objective the original work did not include.

> ***Personal Reflection***
> *This day felt different from the other days. Instead of just summarizing a paper, I noticed a real gap — it never actually tested hallucination — and came up with my own way to fix it. It's still just a proposal, not tested, but it feels more like my own work than the earlier days.*